

**An Investigative Study of Non-Documentary TV Footage as a Data Source for Community
Composition and Biodiversity Research**

Appendix S1

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1. Overview

This README file provides an overview and description of the supplemental information associated with the manuscript titled **An Investigative Study of Non-Documentary TV Footage as a Data Source for Community Composition and Biodiversity Research**. The supplemental file includes a dataset that supports the findings presented in the main text.

2. File List

File Name	Description and Notes	File Type
Appendix S1.pdf	This file – providing context and additional information to support the manuscript as well as background information on other supplementary material files. This file is referred to as Appendix S1 within the manuscript.	PDF
Appendix S2.xlsx	List of identified species and abundance with notes regarding invasive or IUCN red list status.	Excel spreadsheet
Appendix S3.pdf	All the figures and tables that are also in the manuscript with the additional supporting figures and tables that were not presented in the text.	PDF

3. Data Origin, Availability, and Additional Resources

All data were collected by viewing seasons 33-46 of Survivor. All animals that could be identified to a specific family, genus, or species, are listed in the identification excel document. This document lists the organism's common name, scientific name, total abundance observed over the 14 seasons, and the number of different years the taxa was observed. The last column makes note if the taxa is considered invasive or has a status with the IUCN red list. A cleaned-up

database of all observations, including those of animals we could not identify will be made available upon reasonable and agreeable request. Please contact the corresponding author.

Data Structure:

The following variables can be found in the “Appendix S2” excel file:

Variable Name	Description	Unit / Type
Common Name	The common name of the taxa.	Text
Scientific Name	The scientific name of the taxa.	Text
Abundance	Total abundance of the taxa observed over the course of the footage reviewed	Integer
Number of Years	The number of different years that taxa was observed.	Integer
Invasive or IUCN?	Note if the taxa is considered invasive or has a status on the IUCN red list.	Text

Missing Data:

If there is no text noting invasive or IUCN status, then we did not find evidence that particular taxon is considered invasive or had a status on the IUCN red list.

4. Analysis Tools

We conducted our analysis using the following:

- R (version 4.5.0 “How About a Twenty-Six”), with the following packages:
 - Vegan (version 2.7-2) for diversity and species accumulation curve: <https://cran.r-project.org/web/packages/vegan/index.html>
 - Ggplot2 (version 3.5.2) for graphing: <https://cran.r-project.org/web/packages/ggplot2/index.html>
 - Dplyr (version 1.1.4) for data handling: <https://cran.r-project.org/web/packages/dplyr/index.html>
- Operating systems: Windows 11 and macOS Ventura

5. Additional Supporting Text

Additional supporting introductory text regarding the use of various data sources in ecological and conservation research: An increasingly common method to understand temporal shifts in community composition and diversity is the use of historical resources and data, allowing ecologists to generate estimates of diversity and abundance in the past, while enabling comparisons with current conditions. Some of the most common tools in historical ecology are the analysis of historical and archived photographs (Morgan and Gergel 2010, Vellend et al. 2013, Beans 2018, Depauw et al. 2022), preserved specimens used as references (Meineke et al. 2019), as well as old catch/hunting logs and journals (Alexander et al. 2009, Machado et al. 2021). Local or Indigenous peoples may also be consulted or interviewed to document ecological knowledge (Chisholm Hatfield et al. 2018, Azzurro et al. 2019, Kupika et al. 2019). However, these approaches may be limited by barriers to communication and travel, as well as restricted access to technology that facilitates knowledge sharing (e.g., limited or unreliable internet access) or documentation (e.g., lack of cameras to photograph notable events). To overcome these challenges, indirect sources of biological data may be available, whose primary purpose was not to generate ecological information when collected. Data sources like film footage may be valuable.

Additional details regarding selection of *Survivor* in the contexts of date/time: To potentially establish a dataset to allow temporal analysis, we evaluated television programs that filmed in the same locations at generally the same time for several years. The only easily accessible television program that met this requirement, as well as requirements noted in the manuscript, was *Survivor* season 33 to 46. There are more television shows that could be evaluated, but those

programs were not easily accessible to the research team, moved locations consistently, and/or were in locations with organisms outside the author's taxonomic expertise. Additionally, we did not include more television shows due to the amount of time required to collect these data.

Abundance and diversity metrics were grouped by year. Pooling data from two NDTV seasons spanning approximately four months per year reduced short-term variability and random noise, allowing linear models to better detect longer-term trends. We hypothesize that this approach minimizes measurement error and highlights year-to-year changes rather than highly variable episode- or season-level fluctuations. Mathematically, this result is expected, as averaging across multiple observations reduces the influence of extreme values and increases statistical stability.

Additional details regarding resources utilized as identification resources and identification procedures: We utilized our taxonomic expertise in combination with field guides and online resources to support and confirm identification. We took a conservative approach to classification: when species-level identification could not be made with confidence, individuals were classified to the genus level, and when genus-level identification remained uncertain, classification was limited to the family level. If family-level identification was not possible, the classification was retained at one of the 30 previously mentioned taxonomic sub-categories or recorded as "unknown." This approach ensured consistency while minimizing the risk of misidentification. We primarily have backgrounds in tropical and sub-tropical fish, coastal marine invertebrates, and marine mammals. We have experience in bird, terrestrial insects, amphibians, and reptile identification, but have greater expertise and confidence in the previously mentioned marine groups.

Outside of this knowledge, we consulted several resources for identification. These resources included: *Fiji's Natural Heritage* by Paddy Ryan (ISBN-13: 978-0908988143); *Fiji Reef Fish* by Mark Rosenstein (<https://www.fijireeffish.com/>); Avibase – The World Bird Database (<https://avibase.bsc-eoc.org/avibase.jsp?lang=EN>); FishBase (<https://www.fishbase.se/search.php>); The International Union for Conservation's Red List (<https://www.iucnredlist.org/>); and only research grade quality observations from the Mamanuca Islands region on iNaturalist (<https://www.inaturalist.org/>). These resources were used to supplement identifications outside the authors' primary areas of expertise and were most frequently applied to birds and invertebrates. This targeted use of external references helped us improve identification accuracy for taxa that we were less familiar with. With each species identified, we confirmed with multiple sources that the animal's range included the Mamanuca Islands. It should be noted that some species, particularly birds, utilize the region seasonally as a migration stop and/or breeding grounds.

Overall, the animals we observed had distinct coloration markings that greatly aided in narrowing down potential identifications. The exception to this statement were shorebirds with a majority having white and black coloration with finer scale details (e.g., specific black eye-patch shape, black coloration under the wing, or distinct tufts/crests on top of head) required to narrow down the identification. While these characteristics would “normally” be easy to determine, at times, the footage did not allow for the proper angle or image resolution to confirm if these characteristics were absent or present resulting in a less detailed identification. Terrestrial mammals were relatively straightforward to identify. The only native species are bats with only a small number of bat species residing in that specific region aided in possible. All the goats observed are introduced to this island, and the background of their introduction has been

documented in published literature. Amphibians and reptiles could generally be classified to genus based on their distinct physical characteristics and coloration; in contrast, birds and invertebrates were most susceptible to coarser taxonomic classification due to challenges such as small body size or movement (e.g., invertebrates) and distant or aerial observations (e.g., birds). It should also be noted that it is common for tropical and sub-tropical reef fishes to have differing coloration between the earlier life stages (i.e., larval and juvenile) versus the later life stages (i.e., sub-adult and adult), which resulted in extra time to confirm identification.

Additional details regarding verification and evaluation of footage not being used across years:

There are unique landmarks on the specific islands on which Survivor is filmed. Often these landmarks can be clearly observed in the backgrounds of animal footage. Additionally, the set design and items that are unique to the show are also observed. All identified taxa were cross referenced with our identification resources to ensure they are potential residents of the Mamanuca Islands region specifically. Animal composition does change quickly in the region. There are animals present on nearby Viti Levu that are not present on the Mamanuca Islands and vice versa. Lastly, as revealed through press interviews as well as during seasons 49 and 50 (which were not considered in this paper), the crew members of the show also stay on the islands throughout the production of the show. It is highly unlikely that the production would transport crew and equipment to a biodiversity-rich location like the Mamanuca Islands only to capture B-roll footage of wildlife in a different location. With such a popular following, a continuity error such as a mismatch in fauna or setting, would be detected by audience members. For these reasons, we are confident that the animal footage observed was filmed on the Mamanuca Islands.

In terms of not collecting a new set of B-roll each year, we found there were several instances of footage being repeated within the same year, but never across years. From the earlier years to the later years examined, there were multiple changes in image quality in both the “main” footage but also in animal footage. This is likely the contribution of upgraded equipment providing improved image quality. If footage was shared across years, we would have observed animal footage of one image quality against footage of another image quality. We did not observe this. There was also a pattern in the animals observed in the seasons filmed within the same year. When a specific species of animal appeared, it generally appeared multiple times within that year. As an example, Brown Booby (*Sula leucogaster*) or Yellowtailed Fusilier (*Caesio cuning*) both had multiple appearances within a single year, but no appearances outside of that year. If the footage was shared across years, we would expect to see more species that only appeared within a single year to be observed more across years. Overall, 85 out of the 182 taxa were only observed in a single year. If footage was shared across years, we would expect this number to be lower. For these reasons, we are confident that the animal footage observed was collected new each year.

Additional details regarding invasive and IUCN listed species: Of the species identified, there were eight species that have been classified as invasive: Yellow Oriental Paper Wasps (*Pycnonotus cafer*, n=304); Goats (*Capra aegagrus hircus*, n=21); Polynesian Rat (*Rattus exulans*, n=15); Red-vented Bulbul (*Pycnonotus cafer*, n=11); Common House Gecko (*Hemidactylus frenatus*, n=9); Common Myna (*Acridotheres tristis*, n=5); Cane Toad (*Rhinella marina*, n=4); and Coconut Rhinoceros Beetle (*Orcyctes rhinoceros*, n=2).

Additionally, several International Union for Conservation of Nature (IUCN) red listed species were documented in the NDTV footage. There were two data deficient species: Fiji Anchovy (*Sardinella fijiensis*; n=17,080) and Eclipse Butterflyfish (*Chaetodon bennetti*; n=1). There were nine near threatened species including Tawny Hermit Crab (*Coenobita rugosus*; n=83); False Killer Whale (*Pseudorca crassidens*; n=8); Fiji Green Emo Skink (*Emoia concolor*; n=3); Masked Shining Parrot (*Prosopeia personata*; n=3); Crimson Shining Parrot (*Prosopeia splendens*; n=2); Fiji Tree Frog (*Cornufer vitiensis*; n=2); Tridacna Giant Clam (*Tridacna spp.*; n=2); Fiji Bicolor Foxface (*Siganus uspi.*; n=1), and the Fijian Swallowtail (*Papilio schmeltzi*; n=1). Nine vulnerable species were observed: Bull Shark (*Carcharhinus leucas*; n=66); Whitetip Reef Shark (*Triaenodon obesus*; n=42); Blacktip Shark (*Carcharhinus limbatus*; n=33); Manta Ray (*Mobula alfredi*; n=15); Coconut Crab (*Birgus latro*; n=9); Blacktip Reef Shark (*Carcharhinus Melanopterus*; n=5); Fin Whale (*Balaenoptera physalus*; n=2); Indo-Pacific Sergeant Major (*Abudefduf vaigiensis*; n=2), and Tawny Nurse Shark (*Nebrius ferrugineus*; n=1). Three critically endangered species were observed: Hawksbill Sea Turtle (*Eretmochelys imbricata*; n=43); Fiji Petrel (*Pseudobulweria macgillivray*; n=3), and Great Hammerhead (*Sphyrna mokarran*; n=1). Additionally, 55 South Pacific Iguanas (*Brachylophus spp.*) were observed, which are either the endangered Lau Banded Iguana (*Brachylophus fasciatus*) or the critically endangered Fijian Crested Iguana (*Brachylophus vitiensis*). There were 99 taxa of least concern. A full list of those taxa and all other identified taxa are available in supplementary material appendix S2.

Additional supporting text of collaborative data networks and citizen science platforms: NDTV footage data could be combined with or compared against datasets from government or academic

monitoring programs, such as the National Ecological Observatory Network (National Science Foundation 2025) or the Florida Atlantic Coast Telemetry network (Young et al. 2020, FACT Network 2025). Additional platforms, including citizen science applications such as iNaturalist (Campbell et al. 2023, Mason et al. 2025) and BirdNET (Wood et al. 2022), may further enhance species identification and public engagement. Similarly, global trail camera networks and aerial imagery collaborations provide complementary sources of visual biodiversity data (Wingate et al. 2015, Townsend et al. 2021). Pooling NDTV footage with data from established monitoring programs, citizen science platforms, and global imaging networks may help offset selection biases inherent to any single source, enabling broader spatial and temporal analyses.

Additional supporting text of temporal and spatial data analysis: Within NDTV footage, some filming locations exhibited distinct and recurring features, such as rock formations, coastline shapes, or consistent habitat types. These features may offer insight into site-specific changes over time; however, the lack of publicly available metadata limits the ability to assess spatial relationships among filming locations or to link observed changes to specific environmental conditions. This data source could become more valuable and applicable to ecological and conservation research if more temporal and spatial information about the data were available. The ability to apply these data to assess abundance trends and diversity metrics would increase if specific filming data and metadata (e.g., date, time, and location of footage collection) were available. Information on amount of filming effort, location, and timing would allow for better estimation of sampling intensity and better inform researchers of the robustness related to any observed patterns.

Additional supporting text of improvements in filming technology: As camera technology advances, there may be changes in filming equipment, collection protocols, or personnel that can alter the current trends of data collections, the types of animals filmed, and the identification of those animals. This would subsequently influence the rate at which new species are observed and the predicted asymptote of the species' accumulation curve (Figure S4). For example, improvements in camera resolution or shifts in sampling effort and placement may increase species detectability, inflating the estimated asymptote, whereas inconsistent methods or observer bias may reduce detections and lead to underestimation. Consequently, variation in footage collection approaches can influence the number of taxa that could be identified through this data source if datasets are not standardized or if any changes in data collection are not accounted for analytically.

Additional supporting text of the future of NDTV footage in ecology and conservation:

Educational Applications

NDTV footage could serve as an engaging educational tool in zoology courses in which students practice taxonomic identification using video footage with varying degrees of image quality. While not a comparable substitute for in-field identification or practice with preserved specimens, utilization of this footage may provide identification experience similar to imagery collected from underwater monitoring systems (Costa et al. 2006, Harvey et al. 2007) and terrestrial camera traps (Brown and Gehrt 2009, Burton et al. 2015). NDTV footage would also allow for identification practice involving taxa from inaccessible regions and could be used as a supplemental preparation tool prior to field visits or more formal data collection efforts.

Emerging Technologies and Automated Identification

Advances in camera technology may make images clearer, reducing the number of individuals that cannot be identified to any taxonomic group. Artificial intelligence (AI), particularly machine learning-based computer vision models, can automatically identify animal species in footage by detecting patterns in shape, movement, and coloration (Wäldchen and Mäder 2018, Habib et al. 2024). These tools may reduce the time researchers spend manually processing large volumes of video data by rapidly classifying detections at scale. However, outputs from AI systems still require human validation to ensure accuracy and minimize errors, especially for rare, cryptic, or visually similar species. Future research could evaluate the utility and accuracy of these technologies specifically for NDTV footage.

Integration with Existing Monitoring Programs

Future research could directly compare NDTV footage and more established field-collected datasets to determine how the datasets differ in their representation of species composition, abundance, and diversity. The Survivor franchise and related programs, such as Expedition Robinson, include footage from numerous international locations, creating opportunities for NDTV-based analyses across diverse ecological contexts. Pooling NDTV footage with data from established monitoring programs, citizen science platforms, and global imaging networks may help offset selection biases inherent to any single source, enabling broader spatial and temporal analyses. Integration with these complementary datasets may improve the ecological utility of NDTV observations while allowing researchers to evaluate the extent and direction of NDTV-specific biases.

6. Examples of Screen Classifications

There were 5 types of screen classifications. Below we provide links to YouTube videos with notes of examples of each of these screen classifications within the videos. Please note that the videos provided are short clips and/or compilations of footage and do not necessarily reflect the actual footage reviewed. These are just examples.

- https://www.youtube.com/watch?v=xyJDBQmcQ9M&list=PLWwZ4u5_J1L9Ri-WREjUxwrFDXpDBgm0C&index=25
 - i. Unknown flying at 0:23 – background of focus.
 - ii. Unknown flying at 0:38 – background of focus.
 - iii. Flying Fox (*Pteropus spp.*) at 0:41 – scene or frame transition.
 - iv. Unknown Foraging or Shorebird at 1:22 – scene or frame transition. Classified as unknown due to low lighting and movement resulting in inability to get a good look at bird.
 - v. Unknown flying at 2:02 – background of focus.
- https://www.youtube.com/watch?v=p3AU2UJoUng&list=PLWwZ4u5_J1L88Fb8yv0z-oVT1qbCm-BoH&index=6
 - i. Horned Ghost Crab (*Ocypode ceratophthalmus*) at 0:24 – theme song montage.
- <https://www.youtube.com/watch?v=BQtS0I8fGHO>
 - i. Manta Ray (*Mobula alfredi*) and Bluestreak Cleaner Wrasse (*Labroides dimidiatus*) at 0:14 – theme song montage.
 - ii. Manta Ray (*Mobula alfredi*), Sharksucker (*Echeneis naucrates*), unknown reef fish at 0:15 – theme song montage.
 - iii. Tern (*Onychoprion spp.*) at 0:17 – theme song montage.
 - iv. Shorebirds (not classified any further) at 0:18 – theme song montage.
 - v. Tern (*Onychoprion spp.*) at 0:33 – theme song montage.
 - vi. Fiji Anchovy (*Encrasicholina auster*) at 0:42 – theme song montage.
 - vii. Pacific Reef Heron (*Egretta sacra*) at 0:43 – theme song montage.
- <https://www.youtube.com/watch?v=XgDkNazGEu4>
 - i. Earthworm (Family Lumbricidae) at 1:17 – hunted or eaten by cast.
 - ii. Unknown flying at 2:32 – background of focus.
 - iii. Walking stick insect (*Hermarchus apollonius*) at 2:54 – interacting with cast.

7. References listed within this document

- Alexander KE, Leavenworth WB, Cournane J, Cooper AB, Claesson S, Brennan S, Smith G, Rains L, Magness K, Dunn R, Law TK, Gee R, Jeffrey Bolster W, Rosenberg AA. 2009. Gulf of Maine cod in 1861: historical analysis of fishery logbooks, with ecosystem implications. *Fish and Fisheries* 10: 428–449.
- Azzurro E, Bolognini L, Dragičević B, Drakulović D, Dulčić J, Fanelli E, Grati F, Kolutari J, Lipej L, Magaletti E, Marković O, Matić-Skoko S, Mavrič B, Milone N, Joksimović A, Tomanić J, Scarpato A, Tutman P, Vrdoljak D, Zappacosta F. 2019. Detecting the occurrence of indigenous and non-indigenous megafauna through fishermen knowledge: A complementary tool to coastal and port surveys. *Marine Pollution Bulletin* 147: 229–236.
- Beans C. 2018. Journal entries, maps, and photos help ecologists reconstruct ecosystems of the past. *Proceedings of the National Academy of Sciences* 115: 13138–13141.
- Brown J, Gehrt SD. 2009. *The Basics of Using Remote Cameras to Monitor Wildlife*.
- Burton AC, Neilson E, Moreira D, Ladle A, Steenweg R, Fisher JT, Bayne E, Boutin S. 2015. REVIEW: Wildlife camera trapping: a review and recommendations for linking surveys to ecological processes. *Journal of Applied Ecology* 52: 675–685.
- Campbell CJ, Barve V, Belitz MW, Doby JR, White E, Seltzer C, Di Cecco G, Hurlbert AH, Guralnick R. 2023. Identifying the identifiers: How iNaturalist facilitates collaborative, research-relevant data generation and why it matters for biodiversity science. *BioScience* 73: 533–541.
- Chisholm Hatfield S, Marino E, Whyte KP, Dello KD, Mote PW. 2018. Indian time: time, seasonality, and culture in Traditional Ecological Knowledge of climate change. *Ecological Processes* 7: 25.
- Costa C, Loy A, Cataudella S, Davis D, Scardi M. 2006. Extracting fish size using dual underwater cameras. *Aquacultural Engineering* 35: 218–227.
- Depauw L, Blondeel H, De Lombaerde E, De Pauw K, Landuyt D, Lorer E, Vangansbeke P, Vanneste T, Verheyen K, De Frenne P. 2022. The use of photos to investigate ecological change. *Journal of Ecology* 110: 1220–1236.
- FACT Network. 2025. The FACT Network. (<https://secoora.org/fact/>).
- Habib S, Ahmad M, Ul Haq Y, Sana R, Muneer A, Waseem M, Pathan MS, Dev S. 2024. Advancing Taxonomic Classification Through Deep Learning: A Robust Artificial Intelligence Framework for Species Identification Using Natural Images. *IEEE Access* 12: 146718–146732.

- Harvey E, Cappo M, Butler J, Hall N, Kendrick G. 2007. Bait attraction affects the performance of remote underwater video stations in assessment of demersal fish community structure. *Marine Ecology Progress Series* 350: 245–254.
- Kupika OL, Gandiwa E, Nhamo G, Kativu S. 2019. Local Ecological Knowledge on Climate Change and Ecosystem-Based Adaptation Strategies Promote Resilience in the Middle Zambezi Biosphere Reserve, Zimbabwe. *Scientifica* 2019: 1–15.
- Machado AMS, Giehl ELH, Fernandes LP, Ingram SN, Daura-Jorge FG. 2021. Alternative data sources can fill the gaps in data-poor fisheries. *ICES Journal of Marine Science* 78: 1663–1671.
- Mason BM, Mesaglio T, Barratt Heitmann J, Chandler M, Chowdhury S, Gorta SBZ, Grattarola F, Groom Q, Hitchcock C, Hoskins L, Lowe SK, Marquis M, Pernat N, Shirey V, Baasanmunkh S, Callaghan CT. 2025. iNaturalist accelerates biodiversity research. *BioScience* biaf104.
- Meineke EK, Davies TJ, Daru BH, Davis CC. 2019. Biological collections for understanding biodiversity in the Anthropocene. *Philosophical Transactions of the Royal Society B: Biological Sciences* 374: 20170386.
- Morgan JL, Gergel SE. 2010. Quantifying historic landscape heterogeneity from aerial photographs using object-based analysis. *Landscape Ecology* 25: 985–998.
- National Science Foundation. 2025. Neon Science. (<https://www.neonscience.org/>).
- Townsend PA, Clare JDJ, Liu N, Stenglein JL, Anhalt-Depies C, Van Deelen TR, Gilbert NA, Singh A, Martin KJ, Zuckerberg B. 2021. Snapshot Wisconsin: networking community scientists and remote sensing to improve ecological monitoring and management. *Ecological Applications* 31: e02436.
- Vellend M, Brown CD, Kharouba HM, McCune JL, Myers-Smith IH. 2013. Historical ecology: Using unconventional data sources to test for effects of global environmental change. *American Journal of Botany* 100: 1294–1305.
- Wäldchen J, Mäder P. 2018. Machine learning for image based species identification. *Methods in Ecology and Evolution* 9: 2216–2225.
- Wingate L, Ogée J, Cremonese E, Filippa G, Mizunuma T, Migliavacca M, Moisy C, Wilkinson M, Moureaux C, Wohlfahrt G, Hammerle A, Hörtnagl L, Gimeno C, Porcar-Castell A, Galvagno M, Nakaji T, Morison J, Kolle O, Knohl A, Kutsch W, Kolari P, Nikinmaa E, Ibrom A, Gielen B, Eugster W, Balzarolo M, Papale D, Klumpp K, Köstner B, Grünwald T, Joffre R, Ourcival J-M, Hellstrom M, Lindroth A, George C, Longdoz B, Genty B, Levula J, Heinesch B, Sprintsin M, Yakir D, Manise T, Guyon D, Ahrends H, Plaza-Aguilar A, Guan JH, Grace J. 2015. Interpreting canopy development and physiology using a European phenology camera network at flux sites. *Biogeosciences* 12: 5995–6015.

Wood CM, Kahl S, Rahaman A, Klinck H. 2022. The machine learning–powered BirdNET App reduces barriers to global bird research by enabling citizen science participation. *PLOS Biology* 20: e3001670.

Young JM, Bowers ME, Reyier EA, Morley D, Ault ER, Pye JD, Gallagher RM, Ellis RD. 2020. The FACT Network: Philosophy, Evolution, and Management of a Collaborative Coastal Tracking Network. *Marine and Coastal Fisheries* 12: 258–271.